Learning Which Lessons to Show a CTF Agent: Learned Choice Wins on Synthetic Tasks, Not Yet on Real CTFs

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On a synthetic benchmark, using the flag to learn which lessons to show beat a fixed ranking by 13 to 25 points of solve rate. On real CTFs, fixed lessons gave no detectable gain on NYU CTF Bench (+0.6 points [−2.1, +3.4]).

1 The Question

An LLM agent can keep a memory of short **lessons**: notes naming a technique to try. But which lessons should it see for a new challenge?

In capture-the-flag (CTF), an exact-match flag check gives a **verifiable reward** with no human or LLM judge.

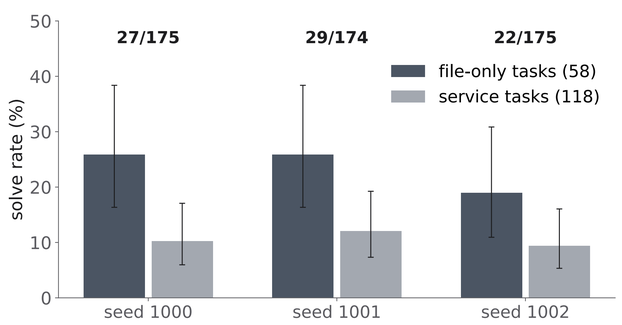
Can learning from the flag pick better lessons than a fixed relevance ranking?

2 Two Test Beds

	Synthetic	NYU CTF Bench
tasks	22 decoding-puzzle families, 2,376 items	176 of 200 test
lessons	48: right + half-right per family, + 4 tool lessons	41, from 45 dev tasks
models	Qwen2.5-14B, -32B	Qwen3.8-27B
tested on	new items, same families	new tasks, 3 seeds

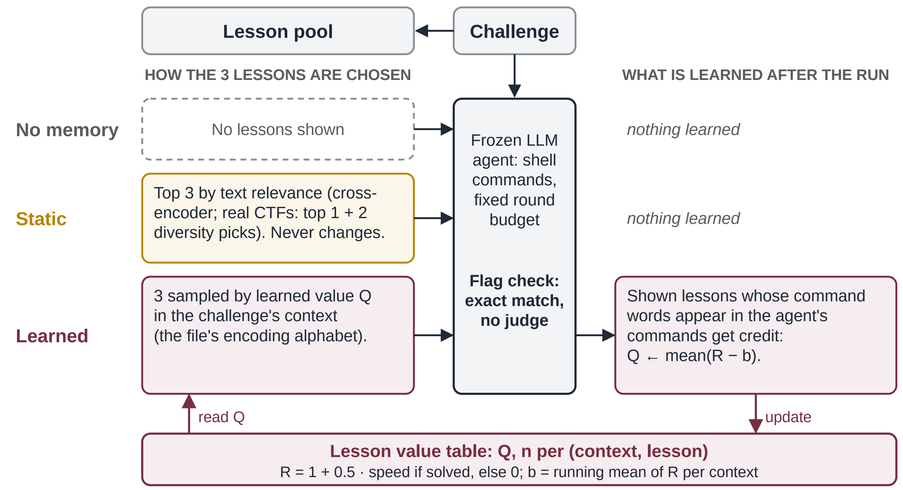
The model writes real-CTF lessons from *other* tasks' files (23 of 41 include a solution). NYU is scored only on tasks the lessons never came from.

3 How Hard Are Real CTFs?



NYU CTF Bench, no memory (Qwen3.8-27B): about 15 % of tasks solved; tasks with a live service are harder. Bars: 95 % Wilson CIs.

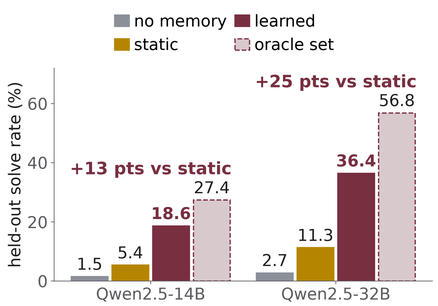
4 How a Lesson Gets Chosen and Credited



Every condition uses the same agent and flag check; only the lesson choice and what is learned from the result differ.

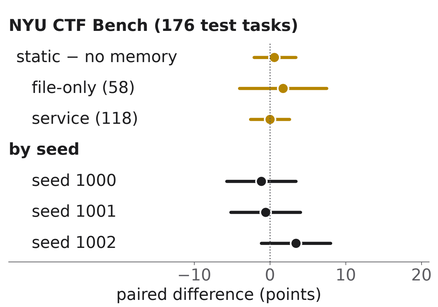
5 Results

Synthetic: learned beat static



New items from the same 22 families; in ∼25 % of training runs, same-family lessons were each shown with probability ½ (same exploration, no context: no gain over static). Oracle set: correct lesson forced in. Causal credit (re-solve without each used lesson) adds 10.6 points over learned at 32B (3 draws, ∼6× the tokens); no detectable gain at 14B.

Real CTFs: static, no detectable gain



Paired by task and seed; task-clustered 95 % CIs. NYU solves: 78 (no memory), 81 (static) of 524 pairs; 25 gained, 22 lost (exact sign test, $p = 0.77$).

6 Key Numbers

+13 and +25 points
learned vs. static, synthetic (14B, 32B; CIs exclude 0)

+0.6 points [−2.1, +3.4]
static vs. no memory, NYU (176 tasks, 3 seeds)

+1.7 points [−4.0, +7.5]
static vs. no memory, NYU file-only tasks

7 Why the Gap? (Hypotheses)

- **Few wins to learn from:** 45 dev tasks at the ∼15 % no-memory rate would give only ∼7 solved runs per pass; the synthetic learners had 510–1,245 in two passes.
- **Coarse credit:** "used" is a keyword match, not proof that a lesson helped.
- **Self-written lessons:** the solver wrote its own lessons from other tasks' files, 18 of 41 with no solution to learn from; a shown lesson appears in its commands in 58 of 524 runs (vs. 46 by chance in the paired no-memory runs), with no detectable change in solve rate.

8 Limits

- Fixed lesson pool; self-extending memory is not tested.
- Real-CTF results cover static lessons only.
- NYU: one model, 3 seeds, 10 rounds of ≤5 commands, internet blocked; 24 test tasks excluded (unsafe, broken, trivial, or non-standard flag format). Public tasks may be in pretraining data.

github.com/jawh3/solver-graph jah21e@fsu.edu